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S/PDIF

S/PDIF (**Sony/Philips Digital Interface**)[1][2] is a type of digital audio interface used in consumer audio equipment to output audio over relatively short distances. The signal is transmitted over either a coaxial cable using RCA or BNC connectors, or a fibre-optic cable using TOSLINK connectors. S/PDIF interconnects components in home theaters and other digital high-fidelity systems.

S/PDIF and TOSLINK connectors on a piece of audio equipment

S/PDIF is based on the AES3 interconnect standard.[3] S/PDIF can carry two channels of uncompressed PCM audio or compressed 5.1 surround sound; it cannot support lossless surround formats that require greater bandwidth.[4]

S/PDIF is a data link layer protocol as well as a set of physical layer specifications for carrying digital audio signals over either optical or electrical cable. The name stands for Sony/Philips Digital Interconnect Format, but is also known as Sony/Philips Digital Interface. Sony and Philips were the primary designers of S/PDIF. S/PDIF is standardized in IEC 60958 as IEC 60958 type II (IEC 958 before 1998).[5]

Applications

A common use is to carry two channels of uncompressed digital audio from a CD player to an amplifying receiver.

The S/PDIF interface is also used to carry compressed digital audio for surround sound as defined by the **IEC 61937** standard. This mode is used to connect the output of a Blu-ray, DVD player or computer, via optical or coax, to a home theatre amplifying receiver that supports Dolby Digital or DTS Digital Surround decoding.

Hardware specifications

S/PDIF was developed at the same time as the main standard, AES3, used to interconnect professional audio equipment in the professional audio field. This resulted from the desire of the various stakeholders to have at least sufficient similarities between the two interfaces to allow the use of the same, or very similar, designs for interfacing ICs.[6] S/PDIF is nearly identical at the protocol level,[a] but uses either coaxial cable (with RCA connectors) or optical fibre (TOSLINK; i.e., JIS F05 or EIAJ optical), both of which cost less than the XLR connection used by AES3. The RCA connectors are typically colour-coded orange to differentiate from other RCA connector uses, such as composite video. S/PDIF uses 75 Ω coaxial cable while AES3 uses 110 Ω balanced twisted pair.

Digital audio coaxial RCA connector (orange)

Signals transmitted over consumer-grade TOSLINK connections are identical in content to those transmitted over coaxial connectors. Optical provides electrical isolation that can help address ground loop issues in systems. The electrical connection can be more robust and supports longer connections.[7]

Comparison of AES3 and S/PDIF[8]

	AES3		S/PDIF	
	Balanced	**Unbalanced**	**Copper**	**Optical**
Cabling	110 Ω STP	75 Ω coaxial	75 Ω coaxial	Optical fibre
Connector	3-pin XLR	BNC	RCA or BNC	TOSLINK
Output level	2–7 V peak to peak	1.0–1.2 V peak to peak	0.5–0.6 V peak to peak	—
Min. input level	0.2 V	0.32 V	0.2 V	—
Max. distance	1000 m	100 m	10 m	
Modulation	Biphase mark code			
Subcode information	ASCII id. text		SCMS copy protection info.	
Audio bit depth	24 bits		20 bits (24 bits, optionally)	

Protocol specifications

S/PDIF is used to transmit digital signals in a number of formats, the most common being the 48 kHz sample rate format (used in Digital Audio Tape and DVDs) and the 44.1 kHz format, used in CD audio. In order to support both sample rates, as well as others that might be needed, the format has no defined bit rate. Instead, the data is sent using biphase mark code, which has either one or two transitions for every bit, allowing the original word clock to be extracted from the signal itself.

S/PDIF protocol differs from AES3 only in the channel status bits; see AES3 § Protocol for the high-level view. Both protocols group 192 samples into an audio block, and transmit one channel status bit per sample, providing one 192-bit *channel status word* per channel per audio block. For S/PDIF, the 192-bit status word is identical between the two channels and is divided into 12 words of 16 bits each, with the first 16 bits being a control code.

S/PDIF control word components[9]

Byte	Bit	Unset (0)	Set (1)
0	0	Consumer (S/PDIF)	Professional (AES3) (changes meaning to AES3 channel status word)
	1	Normal PCM	Compressed data
	2	Copy restrict	Copy permit
	3	2 channels	4 channels
	4	—	—
	5	No pre-emphasis	Pre-emphasis 50/15
	6–7	Mode, defines subsequent bytes; values other than zero are undefined.	
1	0–6	Audio source category indicating the type of source equipment (general, CD-DA, DVD, etc.)	
	7	L-bit, original or copy[A]	
2	0–3	Source number	
	4–7	Channel number	
3	0–3	Sampling frequency: 0000_2: 44.1 kHz, 0100_2: 48 kHz, 1100_2: 32 kHz	
	4–5	Clock accuracy: 10_2: 50ppm,00_2: 1100ppm, 01_2: variable pitch (requires compatible receiver)	
	6–7	Undefined	
4	0	Word length 20 bits	Word length 24 bits
	1–3	Sample length (0: undefined, 1–4: word length minus 1-4 bits, 5: full word length)	
	4–7	Undefined	
5–10	0-7	EAN-13 code (possibly in binary-coded decimal)	
11	0-3		
	4–7	Undefined; padding on 13-digit EAN code	
12–13	0-7	Undefined	
14	0–3		
	4-7	ISRC (encoding unclear; ISRC is 2 alphabetic, 3 alphanumeric and 7 numeric, which is $26^2 \times 36^3 \times 10^7 \approx 2^{48.164}$ and so obviously fits into 7.5 bytes, but a naive 5 ASCII + 7 BCD would be 8.5 bytes)	
15–21	0–7		
22–23	0–7	Undefined	

A. (for most category codes) indicates whether copy-restricted audio is original (may be copied once) or a copy (does not allow recording again). The L-bit is only used if bit 2 is zero, meaning copy-restricted audio. The L-bit polarity depends on the category, with recording allowed if it is 1 for DVD-R and DVD-RW, but 0 for CD-R, CD-RW, and DVD. For plain CD-DA (ordinary nonrecordable CDs), the L-bit is not defined, and recording is prevented by alternating bit 2 at a rate of 4–10 Hz.

Data framing

S/PDIF block contains 192 frames, and each frame contains two sub-frames. A sub-frame has either 20- or 24-bit audio data.[10]

S/PDIF is meant to be used for transmitting 20-bit audio data streams plus other related information. S/PDIF can also transport 24-bit samples by way of four extra bits; however, not all equipment supports this, and these extra bits may be ignored.

To transmit sources with less than 20 bits of sample accuracy, the superfluous bits will be set to zero, and the 4:1–3 bits (sample length) are set accordingly.

IEC 61937 encapsulation

IEC 61937 defines a way to transmit compressed, multi-channel data over S/PDIF.[11]

- The control word bit 0:1 is set to indicate the presence of non-linear-PCM data.
- The sample rate is set to maintain the needed symbol (data) rate. The symbol rate is usually 64 times the sample rate.
- Data is packed into blocks. Each data block is given an IEC 61937 preamble, containing two 16-bit sync words and indicating the state and identity (type, validity, bitstream number, length) of encapsulated data present. Padding is added to match the full block size as required by timing.

A number of encodings are available over IEC 61937, including Dolby AC-3/E-AC-3, Dolby TrueHD, MP3, AAC, ATRAC, DTS, and WMA Pro.[12][13]

Limitations

The receiver does not control the data rate, so it must avoid bit slip by synchronizing its reception with the source clock. Many S/PDIF implementations cannot fully decouple the final signal from the influence of the source or the interconnect. Specifically, the process of clock recovery used to synchronize reception may produce jitter.[14][15][16] If the DAC does not have a stable clock reference, then noise will be introduced into the resulting analog signal. However, receivers can implement various strategies that limit this influence.[16][17]

See also

- ADAT Lightpipe
- I^{2}S
- McASP

Notes

a. Consumer S/PDIF supports the Serial Copy Management System, whereas professional interfaces do not.

References

1. "S/PDIF Information" (https://www.intel.in/content/www/in/en/support/articles/000006034/boards-and-kits/desktop-boards.html). Intel. 21 July 2017. Retrieved 3 April 2018.
2. "S/PDIF" (https://www.pcmag.com/encyclopedia/term/50760/s-pdif). Retrieved 3 April 2018.
3. "SoundSystem SixPack 5.1+ True 6 Channel + Digital In & out – Stuff Worth Knowing" (https://web.archive.org/web/20180702204829/http://terratec.ultron.info/Audio/SiXPack5.1+/Manual/SiXPack5.1+_Manual_GB.pdf) (PDF). TerraTec. 5 July 2001. p. 43. Archived from the original (http://terratec.ultron.info/Audio/SiXPack5.1+/Manual/SiXPack5.1+_Manual_GB.pdf) (PDF) on 2 July 2018. Retrieved 18 January 2011.
4. Mark Johnson; Charles Crawford; Chris Armbrust (2007). *High-Definition DVD Handbook : Producing for HD-DVD and Blu-Ray Disc: Producing for HD-DVD and Blu-Ray Disc* (https://archive.org/details/highdefinitiondv0000john/page/4). McGraw Hill Professional. pp. 4–10 (https://archive.org/details/highdefinitiondv0000john/page/4). ISBN 9780071485852. "...connections such as S/PDIF do not have the bandwidth necessary to deliver uncompressed surround sound..."
5. "Sound card" (http://en.kioskea.net/contents/pc/carte-son.php3). *kioskea.net*. Kioskea Network. Retrieved 4 August 2010. "The components of a sound card are: [...] An SPDIF digital output (Sony Philips Digital Interface, also known as S/PDIF or S-PDIF or IEC 958 or IEC 60958 since 1998). This is an output line that sends digitised audio data to a digital amplifier using a coaxial cable with RCA connectors at the ends."
6. Finger, Robert A. 1992 'AES3-1992: The RevisedTwo-ChannelDigital Audio Interface', J.AudioEng.Soc., Vol.40, No. 3, 1992 March, p108
7. "SPDIF connections: Get connected, not confused" (https://blog.teufelaudio.com/spdif-connections-get-connected-not-confused/). Retrieved 15 May 2024.
8. Dennis Bohn (2001). "Interfacing AES3 & S/PDIF" (http://www.nu9n.com/images/AES3-SPDIF-notes.pdf) (PDF). Rane Corporation. p. 2. Retrieved 18 January 2011.
9. "Understanding/Analyzing Digital Audio Channel Status Bits" (https://web.archive.org/web/20190228070051/https://www.av-iq.com/avcat/images/documents/pdfs/digaudiochannelstatusbits.pdf/). Archived from the original on 28 February 2019. Retrieved 7 July 2022.
10. "Receiving S/PDIF audio stream with the STM32F4/F7/H7 Series" (https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf) (PDF). Retrieved 1 August 2025.
11. Digitalton - Schnittstelle für nichtlinear-PCM-codierte Audio-Bitströme unter Verwendung von IEC 60958 - Teil 1: Allgemeines (IEC 61937-1:2007 + A1:2011); Deutsche Fassung EN 61937-1:2007 + A1:2011
12. "FFmpeg: libavformat/spdif.h File Reference" (https://ffmpeg.org/doxygen/trunk/spdif_8h.html). *ffmpeg.org*.
13. "Representing Formats for IEC 61937 Transmissions - Win32 apps" (https://learn.microsoft.com/en-us/windows/win32/coreaudio/representing-formats-for-iec-61937-transmissions). *learn.microsoft.com*. 15 May 2023.
14. Giorgio Pozzoli. "DIGITabilis: crash course on digital audio interfaces (http://www.tnt-audio.com/clinica/diginterf1_e.html)" tnt-audio.com.
15. Chris Dunn, Malcolm J. Hawksford. "Is the AES/EBU/SPDIF Digital Audio Interface Flawed? (http://www.aes.org/e-lib/browse.cfm?elib=6773)" AES Convention 93, paper 3360.
16. Tracy, Norman. "On Jitter, the S/PDIF Standard, and Audio DACs" (https://web.archive.org/web/20170701142216/http://www.audiocraftersguild.com/AandE/npt.on.jitter2.htm). Archived from the original (http://www.audiocraftersguild.com/AandE/npt.on.jitter2.htm) on 1 July 2017.

17. Lesso, Paul (2006). "A High Performance S/PDIF Receiver" (https://web.archive.org/web/20140604060307/http://www.wolfsonmicro.com/documents/uploads/misc/en/A_high_performance_SPDIF_receiver_Oct_2006.pdf) (PDF). Audio Engineering Society. Archived from the original (http://www.wolfsonmicro.com/documents/uploads/misc/en/A_high_performance_SPDIF_receiver_Oct_2006.pdf) (PDF) on 4 June 2014. `{{cite journal}}`: Cite journal requires `|journal=` (help) AES Convention 121, paper 6948

External links

- S/PDIF at Epanorama.net (http://www.epanorama.net/documents/audio/spdif.html)
- S/PDIF at hardwarebook.net (https://web.archive.org/web/20220911082917/http://www.hardwarebook.info/S/PDIF) at the Wayback Machine (archived 2022-09-11)
- More about channel data bits (http://www.minidisc.org/spdif_c_channel.html)
- Interfacing AES3 and S/PDIF (https://www.ranecommercial.com/legacy/note149.html)